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The Simpson's paradox and fMRI: Similarities and differences between functional connectivity measures derived from within-subject and across-subject correlations

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ABSTRACT

Task-related functional connectivity (*fc-MRI*) indexes the interaction of brain regions during cognitive tasks. Two general classes of methods exist to investigate fc-MRI: the most widely-used method calculates temporal correlations between voxels/regions *within* subjects, and then determines if within-subject correlations are reliable across subjects (*ws-fcMRI*); the other calculates the average (BOLD) signal within voxels/regions and then performs correlations *across* subjects (*as-fcMRI*). That is, while both methods rely on correlational techniques, the level at which correlations are calculated is fundamentally different. While conceptually distinct, it is not known how well these two methods of fc-MRI analyses converge on the same findings. The current study addresses this question across a number of analyses. First, using default-mode network regions as seeds, we show that *as-fcMRI* does not strongly predict *ws-fcMRI* during episodic simulation tasks. Next, we show that the relationship between *as-fcMRI* and *ws-fcMRI* is contingent on whether correlations are calculated between regions from the same functional network (default mode or dorsal attention networks) or between regions from different functional networks. Lastly, we compare seed partial least squares (PLS) – a well-established *as-fcMRI* method – with a novel version of seed PLS that combines the multivariate approach of PLS analyses and within-subject correlations. The results showed that while many regions exhibited congruent *as-fcMRI* and *ws-fcMRI* effects, in some regions the two analyses produced effects in opposite directions. Results are discussed in relation to the Simpson's Paradox, a phenomenon in which across-subject correlations are reversed within individuals present in a sample. Overall, our results suggest that the findings of *as-fcMRI* do not always map onto those from *ws-fcMRI*. We end by discussing the advantages associated with using *ws-fcMRI* to assess the task-related interactions between brain regions.

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1. Introduction

In addition to discerning which brain regions are reliably recruited by certain cognitive processes, functional neuroimaging is increasingly used to characterise the *functional connectivity* of anatomically distinct brain regions (Friston, 2011). This technique assesses the degree of coupling between voxels by correlating the blood oxygenation level dependent (BOLD) signal between them, and can be applied to both resting-state and task-related functional magnetic resonance imaging (fMRI) data (Friston, 2011; Rogers et al., 2007). Resting-state functional connectivity (*rs-fcMRI*) is predominantly determined using a time-series correlation approach, and has been successfully used to elucidate the structure of a number of large-scale brain networks such as the default mode network. Using this approach, the time-series of a seed—often a sphere centred on a voxel of interest (e.g. Fox et al., 2009) or an anatomical region of interest (ROI; e.g. Kim et al., 2011)—is correlated with the time-series in voxels across the brain.

The functional networks elucidated by *rs-fcMRI*, such as the default-mode (DMN) and dorsal attention (DAN) networks, are associated with anatomical connectivity (Greicius et al., 2009) and are conserved both phylogenetically (as shown by their presence in non-human primates: Mantini et al., 2011, 2013) and ontogenetically (as observed in newborn infants: Fransson et al., 2009), suggesting a fundamental organisational principle of neural function. In addition, meta-analytic techniques have shown networks of co-activation across a range of cognitive tasks (e.g. Crossley et al., 2013; Laird et al., 2013; Smith et al., 2009; Toro et al., 2008), which show high degrees of correspondence to the networks observed in *rs-fcMRI* analyses, indicating that the

Abbreviations: as-fcMRI, across-subjects task-related functional connectivity; BOLD, blood oxygen dependent; DAN, dorsal attention network; DMN, default mode network; GM, grey matter; IPS, intraparietal sulcus; PLS, partial least squares; PSC, percent signal change; rs-fcMRI, resting state functional connectivity; ws-fcMRI, within-subjects task-related functional connectivity.

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networks observed at "rest" are largely conserved during experimental tasks (Smith et al., 2009).

Task-related functional connectivity tests whether the interaction between brain regions is modulated by experimental and psychological contexts (Friston, 1994, 2011) and is assessed using a number of distinct correlational or regression approaches. The most important distinction between these approaches is the level of the experimental design at which correlations are computed. Similar to the approach used in *rs-fcMRI* studies, one class of task-related analyses perform *within-subject* correlations on time-series data extracted from seed regions. Examples of such task-related functional connectivity analyses are psychophysiological interactions[1] (PPI: Friston, 1994, 2011) and beta-series correlations (Rissman et al., 2004). While these analysis techniques differ in their underlying computations, and can produce different results (Cisler et al., 2014; McLaren et al., 2012), they nonetheless rely on the relationship between time-series from different regions, and are thus motivated by a definition that regards functional connectivity as "the *temporal* correlation(s) between remote neurophysiological events" (Friston, 1994—our emphasis).

However, task-related functional connectivity is also assessed in some studies by performing *across-subject* correlations independently of any temporal information. Essentially, these studies first determine the average activation of voxels within conditions for each subject (thus averaging across the time-series and discarding any temporal information), and use these values to calculate *across-subject* correlations. Pairs of voxels exhibiting high correlation coefficients are regarded as being functionally connected. Such analyses were originally performed on PET data, which has particularly poor temporal resolution (e.g. Horwitz et al., 1998; Zald et al., 1998). This rationale has, however, also been adopted in some fMRI studies despite the acquisition of temporally-laden data (e.g. Damaraju et al., 2009; Poldrack et al., 2001). Finally, a multivariate analysis procedure called seed partial least squares (seed PLS) performs across-subject correlations of seed region activity with the rest of the brain, and then determines whether these correlations differ between seed regions, experimental conditions or groups[2] (McIntosh et al., 1996, 1997; McIntosh and Lobaugh, 2004).

Given that these two families of analyses (temporal within-subject correlations and atemporal across-subject correlations) share a common name (i.e., "functional connectivity"), it is tempting to assume that *across*-subject correlations between two regions inform our understanding of connectivity between those regions *within* individual brains in the sample (and vice-versa). Strother et al. (1995) have suggested that the structure of intra- and inter-subject covariance matrices (i.e. within- and across-subject functional connectivity) are unlikely to differ substantially, as such a difference would require across-subject variability to be related to cognitive and functional processes distinct from those associated with within-subject variability. However, as noted by others (e.g. Friston, 1995; Horwitz, 2003), it is not at all clear that these two methods of calculating functional connectivity—within- and across-subject correlations—capture a singular phenomenon or produce covariance matrices with similar structures. Pertinent to this issue is the Simpson's Paradox, which occurs when some statistical effect present at a group-level analysis is reversed within individuals in that group (Kievit et al., 2013). For example, in the field of psychological individual differences, Hamaker et al. (2007) have shown that the inter-individual relationship between extraversion and neuroticism is not mirrored within individuals over time. The relevance of these findings to how one measures functional connectivity is clear: it is, for example, possible for the time-series of two voxels to be negatively correlated in every individual in a sample, but for the mean BOLD signal of the two voxels to be positively correlated across subjects (as depicted in Fig. 1).

Across-subject effects are only *necessarily* mirrored within subjects when the process under investigation meets the very strict conditions of ergodicity (Molenaar, 2008; Molenaar and Campbell, 2009). Briefly, any neural or psychological process is ergodic if two conditions are met. First, the process has to be stationary, such that the statistical parameters associated with that process are constant over time within individuals. In the case of functional connectivity, this requires the coupling of brain regions to be approximately constant over time, which is highly unlikely to be true (Leonardi et al., 2013). Second, ergodic processes are homogenous across individuals within a sample. In relation to functional connectivity, this homogeneity assumption requires that the temporal correlations between a pair of brain regions are the same across subjects. Given these conditions, it is questionable that neural processes are ergodic (Masuda and Aihara, 2003; Medaglia et al., 2011). This is not to say that two regions cannot, in principle, reliably exhibit both across- and within-subject correlations. Instead, it merely highlights the point that there is no *necessary* congruence between these two methods of calculating functional connectivity.

The current study seeks to clarify the degree to which across-subject and within-subject correlations converge on the same organisational principle: "functional connectivity". That is, do regions exhibiting within-subject functional connectivity (*ws-fcMRI*) also exhibit across-subject functional connectivity (*as-fcMRI*), and are the spatial patterns of connectivity revealed by both methods similar? This is an important question because many studies using *as-fcMRI* as an index of task-related functional connectivity interpret any significant across-subject correlations between regions from distinct resting-state functional networks as reflecting task-induced reconfigurations of resting-state functional networks. For example, Spreng and Grady (2010) used seed PLS to investigate whether a seed from the DMN (medial prefrontal cortex) and another from the DAN (intraparietal sulcus, IPS) exhibit differential functional connectivity during a range of tasks that activate the DMN. They showed that the IPS seed exhibited reliable *as-fcMRI* with regions in the DMN, and this observation was interpreted as the cognitive tasks inducing a shift in IPS connectivity from the DAN—as is commonly observed in resting-state studies (Fox & Raichle, 2007; Q6 Yeo et al., 2011)—to the DMN. The implicit assumption is that Q7 resting-state and seed PLS analyses are both tapping into the same underlying phenomenon (functional connectivity), and the most relevant distinction between the two approaches is that only the latter is task-related—ignoring the fact that these methods also fundamentally differ in the level at which correlations are computed.

To investigate this question, fMRI data from a study investigating episodic memory and future thinking (Hach et al., 2014) were submitted to a series of analyses. Importantly, the main concern here is to assess similarities and differences between functional connectivity methods, and not between the experimental conditions per se. First, DMN regions engaged during these tasks (Schacter et al., 2007) were selected as seeds. Regression analyses were performed to assess how well seed-to-voxel across-subject correlations predict within-subject correlations across the brain. Next, correlation matrices were created using *as-fcMRI* and *ws-fcMRI* methods to assess if the relationship between these two forms of functional connectivity was influenced by whether the correlations were calculated for regions belonging to the same functional network (i.e., DMN and DAN) or between regions from each of these networks. Finally, two seed regions from each of these networks were used in two separate PLS analyses: a standard seed PLS analysis employing across-subject correlations (*as-seed PLS*) and a modified version of seed PLS based on within-subject correlations (*ws-seed PLS*); spatial patterns were produced for each analysis allowing a comparison of these two forms of functional connectivity.

[1] The classification of PPI is not straightforward. While many authors refer to it as a form of effective connectivity (Friston, 2011), it is classified by others as a form of functional connectivity (O'Reilly et al., 2012). The distinction between these two classifications is beyond the scope of the current paper.

[2] It is important to note that event-related seed PLS analyses, which make use of temporal information in the form of the *average* response of voxels over time, do not index within-subject, temporal correlations between voxels. Instead, this method calculates across-subject correlations at different TRs within a specified temporal window.

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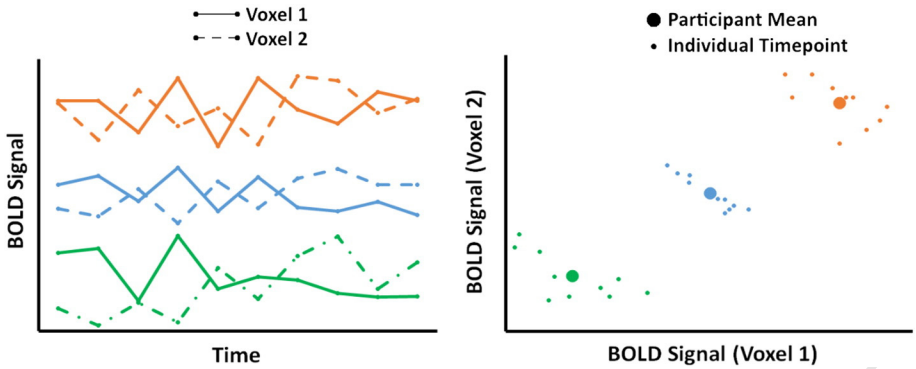

Fig. 1. Hypothetical scenario showing, on the left, a negative relationship of BOLD signal over time for two voxels in three participants (shown in different colours) and, on the right, a positive across-subject relationship of the voxel's participant means. BOLD = blood oxygenation level dependent. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2. Methods

2.1. Participants

Sixteen right-handed participants (2 males) between the ages of 18–28 years (M = 21 years; SD = 3) took part in the study. Participants had no history of traumatic brain injury, neurological conditions, alcohol or substance dependency or psychiatric illness. The study was approved by the local ethics committee (University of Auckland Human Participants Ethics Committee) and participants provided informed consent prior to commencement of testing.

2.2. Tasks

Three experimental conditions were presented in the scanner over six 7-minute runs in a slow-event related design: episodic memory of past events, imagination of future events and a control semantic task.[3] Briefly, in 24 *Past* and 24 *Future* trials, participants were presented with an event-type cue (e.g., "family celebration") and were required to recall a past event or imagine a future event related to the cue. In 24 *Control* trials, participants were shown a cue word (noun) and required to (1) retrieve two objects semantically related to the cue word, (2) arrange all three words (i.e., cue word and two retrieved words) by physical size from largest to smallest, and (3) define the objects. Each trial lasted 20 s, after which participants were given 10 s to make a set of ratings for the trial (e.g., detail generated; task difficulty). Trials were randomly ordered and separated by jittered fixation trials (mean duration, 5 s; range, 3.5–6.5 s).

2.3. Data acquisition

Images were acquired on a 3 Tesla Siemens Skyra MRI scanner with a 20-channel receiver coil. Detailed anatomical data were collected using a multiplanar rapidly acquired gradient echo (MP-RAGE) sequence. Functional images were acquired using a T2*-weighted echo planar imaging (EPI) sequence (TR = 3000 ms, TE = 27 ms, FOV = 225 mm, flip angle = 90°). Fifty-nine coronal oblique slices (3.5 mm thick, 0.7 mm gap) were acquired at an angle perpendicular to the long axis of the hippocampus in an interleaved fashion. For distortion correction (Hutton et al., 2002), field maps were acquired with a standard manufacturer's double echo field map sequence (TE = 4.92 and 7.38 ms, TR = 577 ms) using 59 slices covering the whole head.

2.4. Data preprocessing

To allow the longitudinal magnetisation to reach equilibrium, the initial four images from each run were discarded. Pre-processing of functional images was performed using SPM8 (Statistical Parametric Mapping, Wellcome Trust Centre for Neuroimaging, Institute of Neurology, London, UK) run in Matlab (version 8.0.0.783, Mathworks Inc., MA) and included slice timing correction, adjustment for movement across scans through realignment and unwarping, co-registration with the anatomical scan, spatial normalisation (using normalisation parameters derived during segmentation) to the Montreal Neurological Institute (MNI) template (resampled at 2 mm × 2 mm × 2 mm), and spatial smoothing (using an 8 mm full-width half maximum isotropic Gaussian kernel).

To remove data artefacts, independent components analysis (ICA) was performed (using FSL's Melodic utility, version 3.14) and components manually classified as structured noise were removed using procedures described by Kelly et al. (2010). In addition, non-neural sources of variance were accounted for by regressing out motion parameters and white-matter and CSF signal from each voxel's time-course. A custom MNI grey-matter (GM) mask was applied to ensure only BOLD signal from GM tissue was included in the analyses.

2.5. Data analysis: Overview of methods

2.5.1. Partial least squares

Task PLS is a multivariate technique that identifies latent variables correlating whole brain patterns of activity with an experimental design (Krishnan et al., 2011; McIntosh et al., 1996, 2004). Seed PLS is a variant that, instead of examining patterns of activation, assesses *as-fcMRI* by correlating the mean BOLD signal in an ROI with all voxels in the brain for each experimental condition. As described in more detail by McIntosh and Lobaugh (2004) and Krishnan et al. (2011), first the cross product (**R**) of two matrices, $\boldsymbol{X}$ and $\boldsymbol{Y}$, is computed (i.e., $\boldsymbol{R} = \boldsymbol{X} * \boldsymbol{Y}$). $\boldsymbol{X}$ is a matrix containing a measure of brain activity (mean percent signal change, PSC)[4,5] from each voxel in the brain. $\boldsymbol{Y}$ is a matrix that, for task PLS, contains a set of contrasts specifying the experimental design; for seed PLS, $\boldsymbol{Y}$ contains vectors of seed activity during each condition (thereby also coding for experimental design).

Matrix $\boldsymbol{R}$ is submitted to a singular value decomposition, resulting in latent variables (LVs) that capture the maximal covariance between matrices $\boldsymbol{X}$ and $\boldsymbol{Y}$. Each LV is defined by (i) a singular value indicating

[3] For a more detailed description of the experimental tasks, see Hach et al. (2014).

[4] Note that in PLS, the BOLD signal is normalised to the first TR of each trial; thus PSC values reflect mean change relative to the start of trials rather than an implicit baseline.

[5] Matrix $\boldsymbol{X}$ is not restricted to mean PSC; recent studies have used task PLS to investigate intra-individual BOLD variability in which $\boldsymbol{X}$ contains standard deviations of BOLD signal (Garrett et al., 2010, 2011, 2013).

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the amount of covariance for which the LV accounts; (ii) a linear contrast across conditions and groups that codes the effect depicted by voxels; and (iii) a singular image of voxel weights or "saliences" that displays the brain regions exhibiting the greatest covariance with the linear contrast. Brain scores—the dot product of voxel saliences and PSC values for all voxels in each condition—indexes how strongly each participant expresses the spatial pattern of brain activity associated with each condition in a given LV. In seed PLS, brain scores provide a weighted average of BOLD signal in regions showing reliable across-subject correlations with a seed region.[6]

To assess the significance of LVs, permutation tests are performed whereby brain data (task PLS) or seed values (seed PLS) are randomly reassigned to experimental conditions for each subject. The permuted matrix is again submitted to singular value decomposition to produce a set of LVs. This procedure is repeated to create a sampling distribution of singular values under the null hypothesis; if the observed singular value is larger than 95% of these permuted singular values, the LV is regarded as significant at a threshold of $p < .05$. In addition, the reliability of each voxel's salience is determined by first estimating the standard error (SE) using a bootstrap procedure: subjects are randomly resampled with replacement and the PLS analysis is re-run to derive new saliences for each voxel. After running this process repeatedly, the SE of the derived saliences is computed. A bootstrap ratio (original salience/SE of derived saliences) is then calculated for each voxel, with bootstrap ratios of >2 (approximately equivalent to $p < .05$) regarded as reliable. Confidence intervals (95%) for the mean brain scores in each condition are also calculated from the bootstrap procedure. Examination of confidence intervals indicates how reliably each condition/seed contributes to the spatiotemporal pattern associated with the contrast (confidence intervals crossing zero indicate that a condition/seed did not contribute reliably to the pattern). Importantly, no correction for multiple comparisons is necessary; because of its multivariate nature, PLS involves only a single analytic step.

2.5.2. Calculation of within and across-subject correlation coefficients

In the current study, analyses involved direct comparisons between *ws-fcMRI* and *as-fcMRI*; these two forms of correlations were computed in the following way. As shown in Fig. 2, for *ws-fcMRI*, a vector of PSC values (i.e., one value for each trial) was extracted, creating a trial-series of values for each voxel in each condition.[7] PSC was calculated relative to the first TR of each trial. For seed regions, PSC values from GM voxels immediately surrounding a central peak voxel (i.e., one-nearest neighbours, resulting in a 12 mm cube) were averaged for each *trial* producing a vector of PSC values (i.e., a seed trial-series). Within-subject correlations were then performed between each seed trial-series and the trial-series in every voxel, yielding a vector of r values (with a length corresponding to the number of voxels in the GM mask) for each condition, specifying the within-subject correlation between the seed and every GM voxel.[8] This process was performed separately for each participant.

For *as-fcMRI*, mean PSC at each voxel was calculated for each condition in each participant by averaging across all trials. Values for GM voxels within each seed region were averaged to produce one mean PSC value per seed per participant. Correlations of mean PSC data were then performed (separately for each condition) between each seed region and all GM voxels in an across-subject fashion, resulting in a vector (with a length corresponding to the number of voxels in the GM mask) of r values indexing the across-subject correlation each voxel correlates with the seed regions.

2.5.3. Selection of seed regions

A non-rotated task PLS analysis was run to identify regions significantly engaged by the *Past* and *Future* tasks relative to the *Control* task. Each trial in the slow event-related design was treated as a mini-block 7 TRs in duration (21 s), approximating the 20 s remember/imagination phase of each 30 s trial. For this analysis, 1000 permutation tests and 1000 bootstraps were run. Voxels in which bootstrap ratios were greater than 5 (roughly equal to a p value of <.0001) with a minimum cluster size of 5 voxels were considered reliable. MNI coordinates of peak voxels from each cluster (i.e., the voxel with the highest bootstrap ratio) were localised according to Harvard-Oxford cortical and subcortical atlases. Peak voxels from eight significant clusters from this task PLS analysis were used to define DMN seed regions for subsequent functional connectivity analyses. All seeds from the task PLS overlapped with a resting-state DMN mask acquired from www.neurosynth.com (Yarkoni et al., 2011). In addition, we selected eight peak voxels within the DAN as defined by Yeo et al. (2011). The list of peak voxels for seed regions is presented in Table 1. Q9

2.6. Functional connectivity data analysis

2.6.1. Whole-brain regression analyses

In our first set of analyses, we sought to determine whether, for DMN seeds, across-subject correlations can reliably predict mean within-subject correlations across voxels. To this end, we used the across-subject and within-subject r-values computed for each condition (see Section 2.5.2); within-subject r-values were further averaged across participants generating for each seed/condition a set of mean r-values that indexed the average within-subject correlations between the seed and every voxel during that condition. For each DMN seed, a linear regression analysis was performed, across voxels, assessing the relationship between the two sets of correlation coefficients for both conditions.

2.6.2. ws-fcMRI, as-fcMRI and functional networks

While the first set of analyses was aimed at determining the relationship between the two forms of functional connectivity between a seed and all voxels in the brain, we restricted this next set of analyses to a subset of regions within the DMN and DAN. The rationale behind this approach was to determine if the relationship between *ws-fcMRI* and *as-fcMRI* is influenced by whether the two regions between which functional connectivity is measured belong to the same functional network or are from different brain networks. For each condition, either across-subject r-values (Section 2.5.2) or mean within-subject r-values (Section 2.6.1) from the eight DMN and eight DAN seeds were used to create correlation matrices. In this way, we could determine whether both *as-fcMRI* and *ws-fcMRI* can demarcate the two functional networks. Moreover, using the linear regression approach detailed in Section 2.6.1, we assessed whether the ability of *as-fcMRI* to predict *ws-fcMRI* differed for within-network and across-network correlations.

2.6.3. Seed PLS analyses

In this set of analyses, we used PLS to assess the overlap (or lack thereof) of the connectivity derived by *as-fcMRI* and *ws-fcMRI* methods. Two seeds – each a key node of their respective functional networks (Fox et al., 2005; Utevsky et al., 2014) – were selected for the PLS analyses: the right precuneus (DMN) and left IPS (DAN; see Table 1 for peak coordinates). To examine whole-brain patterns of *as-fcMRI*, a non-rotated version of seed PLS (as-seed PLS) was performed using a contrast that identified voxels exhibiting significant differential across-

[6] While brain scores in task PLS index how strongly a participant is showing an effect based on BOLD signal differences, brain scores in seed PLS do not indicate how strongly a participant is exhibiting some functional connectivity effect (even though seed PLS is a measure of functional connectivity). That is, brain scores in task PLS map onto what the analysis is measuring (task induced changes in BOLD signal), but this is not the case for seed PLS.

[7] This approach is conceptually the same as beta-series correlations (Rissman et al., 2004) in that the unit of time relevant to this form of functional connectivity analysis is the trial (as opposed to TRs in a typical time-series analysis).

[8] Concatenating trials and then generating r-values using TRs (instead of trials) as the unit of time when calculating temporal correlations produced highly similar results to those reported here.

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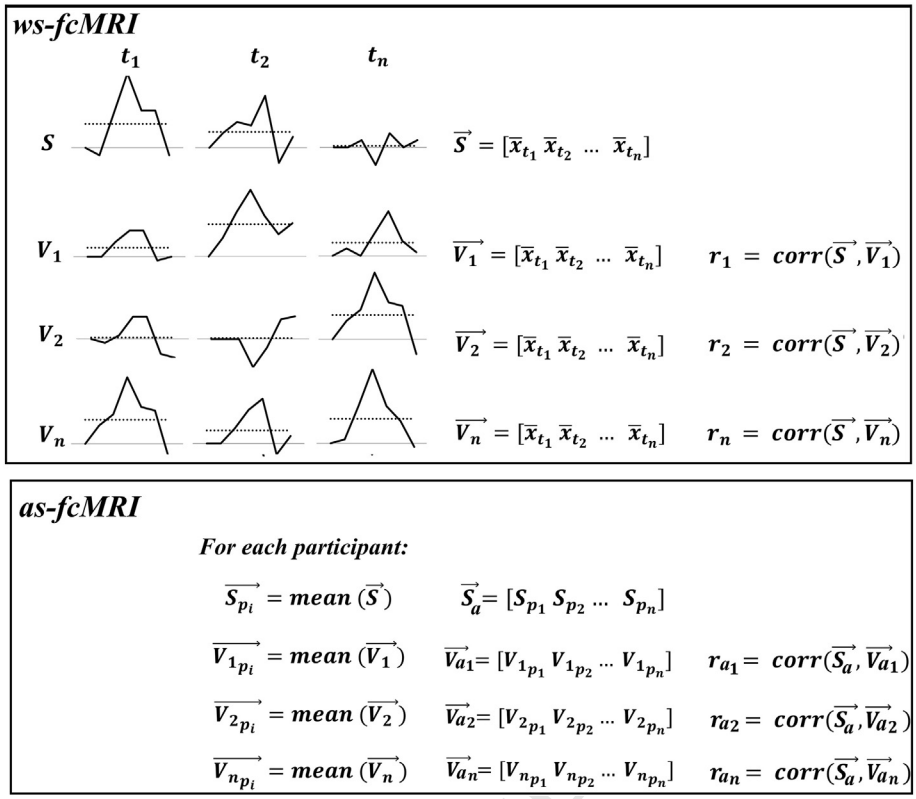

Fig. 2. A schematic showing the method of calculating within- and across subject *r*-values. Upper panel: For within-subject r-values (*ws-fcMRI*), mean PSC ($\boldsymbol{x_i}$: dotted lines) was calculated for each trial ($\boldsymbol{t_i}$) for both the seed region ($\boldsymbol{S}$) and each voxel ($\boldsymbol{V_i}$). Note, for the seed region, PSC is the mean of voxels within a 12 mm cube surrounding a seed voxel. This procedure resulted in vectors for the seed region ($\vec{S}$) and every voxel ($\vec{V_i}$). Pearson correlation coefficients ($\boldsymbol{r_i}$) specifying the strength of the relationship between $\vec{S}$ and each voxel were then calculated for each participant. Lower panel: For across-subject *r*-values (*as-fcMRI*), mean PSC (averaged across trials) for the seed region ($\vec{S_{p_i}}$) and each voxel ($\boldsymbol{V_{i_{pi}}}$) were first calculated for each participant. This procedure resulted in one vector of participant mean PSC values for the seed region ($\vec{S_a}$) and for each voxel ($\vec{V_{ai}}$). Pearson correlation coefficients ($\boldsymbol{r_{ai}}$) were computed specifying the across-subject relationship between the seed region and every voxel.

subject correlations with the seeds, collapsed across both conditions (i.e. a main effect of seed). A multi-seed analytic approach was adopted to determine, for each functional connectivity method, how strongly each seed covaried with its functional network. This approach was motivated by previous research showing that key nodes of a given resting state functional network can exhibit reliable *as-fcMRI* with networks that are anti-correlated with the seed during rest (e.g. Grady et al., 2010; Protzner and McIntosh, 2009; Spreng and Grady, 2010). In order to directly compare *ws-fcMRI* with *as-fcMRI* within the same multivariate framework, we developed a novel seed PLS procedure to examine whole brain patterns of *within-subject* correlations between seed regions and voxels. To identify voxels exhibiting reliable *ws-fcMRI* with the two seed regions, a non-rotated task PLS was performed on within-subject *r*-values that treated each seed-condition combination as a separate condition (i.e., Precuneus-Past; Precuneus-Future; IPS-Past; IPS-Future). To clarify, while *task* PLS analyses are generally performed on mean PSC values to identify regions in which mean BOLD signal covaries with experimental conditions (McIntosh and Lobaugh, 2004), our analysis was performed on within-subject *r*-values to assess which regions showed reliable *ws-fcMRI* with the seed regions during experimental conditions.

For both PLS analyses, 1000 permutation tests and 1000 bootstrap procedures were run and clusters of 5 or more voxels in which bootstrap ratios were greater than 2.0 (roughly equal to a *p* value of .05) were considered to represent reliable voxels. MNI coordinates of peak voxels from each cluster (i.e., the voxel with the highest bootstrap ratio) were localised according to Harvard-Oxford anatomical atlases. The spatial maps from these two analyses were then compared to highlight regions in which the direction of a significant *as-fcMRI* effect was the same as or opposite to that derived using *ws-fcMRI*.

Table 1
List of seed regions used in functional connectivity analyses.

Structure	MNI co-ordinates x	y	z
DMN regions			
L frontal pole/medial prefrontal cortex	−4	56	−8
R precuneus	4	−56	12
L angular gyrus	−48	−68	20
R angular gyrus	48	−64	20
L middle temporal gyrus (temporal pole)	−52	−12	−16
R middle temporal gyrus (temporal pole)	64	−12	−24
R parahippocampal gyrus/hippocampus	24	−28	−24
L hippocampus/parahippocampal gyrus	−24	−20	−24
DAN regions			
B Intraparietal sulcus	±36	−40	44
B Supplementary Motor Area	±40	48	4
B Superior Parietal Lobule	±20	−68	52
B Area MT	±52	−64	−4

Note. B = bilateral; DMN = default mode network; DAN = dorsal attention network; L = left; MNI = Montreal Neurological Institute; R = right. DMN regions were taken from a Future and Past > Control task-PLS contrast, and DAN seeds were taken Yeo et al. (2011).

3. Results

3.1. Whole-brain regression analyses

Linear regression analyses were performed, across voxels, to determine whether across-subject correlations could predict mean within-subject *r*-values for each of the seed regions for the two experimental conditions. As shown in Fig. 3, while the relationships between these

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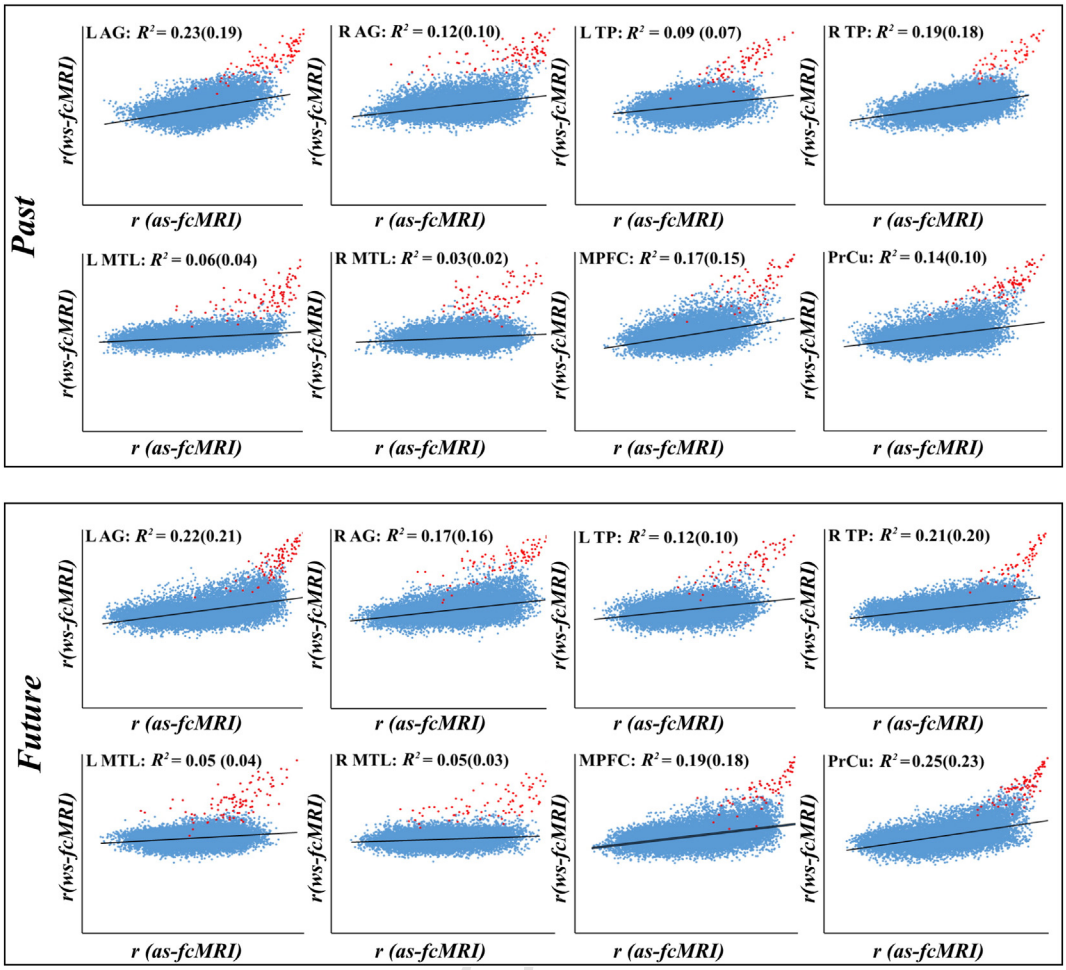

Fig. 3. Scatterplots showing the relationship between *as-fcMRI* and mean *ws-fcMRI* for all voxels in the brain for selected DMN seeds. The top panel is for *Past* and the lower panel for *Future* conditions. Red dots indicate voxels within 4 mm of the seed region; blue dots indicate voxels in the rest of the brain. R^2 values indicate how strongly as-fcMRI predicts *ws-fcMRI*. R^2 values were calculated across all voxels as well as when voxels immediately surrounding the seed region (red dots) were excluded. These R^2 values are presented in parentheses.

two measures of functional connectivity were significant for all of the seeds, the effects were nonetheless weak to moderate (R^2 values ranged from .03 to .25). When the voxels immediately surrounding the seed region (i.e., within 4 mm of the perimeter of the seed region) were excluded from the analysis, such that only more distant voxels were considered, these relationships weakened.

3.2. Correlation matrices in the default mode and dorsal attention networks

The next set of analyses aimed to determine whether the relationship between *ws-fcMRI* and *as-fcMRI* was influenced by whether the two regions between which functional connectivity is measured belong to the same or different functional networks. That is, our interest here is not in the correlations between seed regions and all voxels in the brain, but between the seed regions themselves — half of which were located in the DMN, the other half in the DAN. To this end, we compared the correlation matrices produced by each functional connectivity method to determine how well the two methods aligned with respect to connectivity within and across the DMN and DAN. These matrices contained correlations specifying the relationship between the selected seed regions, as opposed to the relationship between seed regions and all voxels in the brain.

Visual inspection of these correlation matrices (shown in Fig. 4) suggest that both forms of functional connectivity can distinguish between the DMN and DAN, with correlations within these networks (the top left and bottom right quadrants of each correlation matrix) generally being more positive relative to correlations across networks (bottom left/top right quadrants). There are, however, two differences between *as-fcMRI* and *ws-fcMRI* correlation matrices. First, across-subject correlations produce more variable *r*-values, especially for correlations between nodes in different networks. In addition, across-subject *r*-values produce substantially more negative correlations than within-subject *r*-values. While this latter pattern is more apparent between nodes of different networks, there are also a number of negative correlations between nodes of the same network.

For *as-fcMRI*, the significance of each functional connection was assessed, showing that significant positive correlations ($p < .05$, uncorrected) were more likely to occur for regions within the same network than between regions from different networks. Conversely, significant negative correlations generally occurred for regions from different networks, although this effect was only apparent for connections to the right area MT during the *Future* condition. For *ws-fcMRI*, the significance of each correlation was assessed by first performing an *r*-to-*z* transformation of *r*-values for each participant. One-sample *t*-tests were then performed to determine if the strength of the relationship between each pair of regions was reliable (i.e., different from zero). These analyses showed that, in a similar vein to the *as-fcMRI* analyses, significant positive correlations were more likely to occur between regions from the same network than regions between different networks. However, it should be noted that the number of significant within-network connections was much greater for the *ws-fcMRI* approach than the *as-fcMRI* approach: of the 56 within-network relationships (28 for each condition), 49 were significant for *ws-fcMRI* while 12 were so for the *as-fcMRI* approach.

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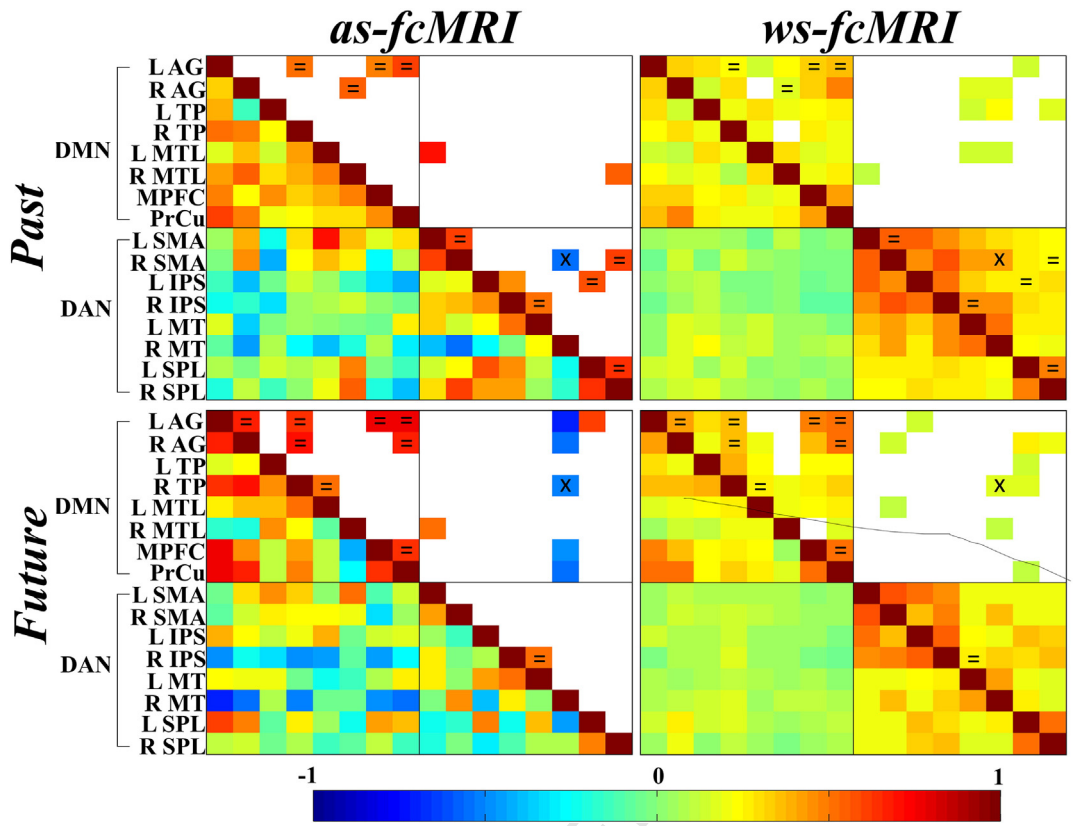

Fig. 4. Correlation matrices showing functional connectivity between regions in the DMN and DAN for Past and Future conditions using *as-fcMRI* and *ws-fcMRI* as methods of calculating correlations between regions. For each matrix, only significant correlations are reproduced above the main diagonal. Correlations that are significant for both methods and going in the same direction are denoted by "=". Correlations that are significant and going in the opposite direction are denoted by "X". The colour bar represents *r*-values. AG = angular gyrus; *as-fcMRI* = across-subject task-related functional connectivity; DAN = dorsal attention network; DMN = default-mode network; IPS = intraparietal sulcus; L = left; MPFC = medial prefrontal cortex; MT = middle temporal area; MTL = medial temporal lobe; PrCu = precuneus; R = right; SMA = supplementary motor area; SPL = superior parietal lobule; TP = temporal pole; *ws-fcMRI* = within-subject task-related functional connectivity.

To interrogate the similarities and differences between the correlation matrices in more detail, two subsequent analyses were performed. First, to assess whether each form of functional connectivity could reliably distinguish between DMN and DAN, a bootstrapping procedure was used to determine 95% confidence intervals of the mean *r*-to-*z* transformed correlation coefficient between regions within each functional network, as well as between regions from different networks. To avoid duplicates and infinite *z*-scores, only correlation coefficients below the main diagonal were used to calculate means. This procedure was completed for both *as-fcMRI* and *ws-fcMRI* correlation matrices. If the two forms of functional connectivity analyses reliably demarcate the two functional networks, it should be the case that the mean *z*-scores within each network are significantly greater than *z*-scores between the networks (i.e., the 95% confidence intervals should not overlap). Given that we were primarily concerned with similarities and differences between *as-fcMRI* and *ws-fcMRI*, correlation coefficients from the two experimental conditions were averaged.

A lack of overlap of 95% confidence intervals showed that correlation coefficients were significantly greater for correlations between regions in the same network relative to correlations between regions from different networks for both measures of functional connectivity, indicating that both measures are able to demarcate the DMN and DAN.

Next, we assessed the linear relationship between the two sets of correlation coefficients to assess two questions: (1) Can across-subject correlations predict the degree of within-subject functional connectivity between regions in the correlation matrices?; and (2) Does the strength of this prediction differ depending on whether the correlations are from within the same network? As seen in Fig. 5, across-subject correlations strongly predicted within-subject correlations between regions within the DMN ($R^2 = .47$, $p < .001$). This relationship was weaker – but significant – between regions within the DAN ($R^2 = .10$, $p = .016$). There was, however, no such relationship for correlation coefficients between regions from two different functional networks ($R^2 = .00$, $p = .48$).

Finally, we assessed whether the relationship between any pairs of regions exhibited a Simpson's Paradox effect. This was done by identifying whether any pairs of regions showed a significant across-subject correlation (i.e. $p < .05$) and a statistically reliable within-subject correlation in the *opposite* direction. Such reversals were evident for the relationship between the right SMA and right area MT during the *Past* condition as well as between the right temporal pole and the right area MT during the *Future* condition. In both cases, a significant negative *as-fcMRI* relationship between the pair of regions is contrasted with a significant positive *ws-fcMRI* relationship. A graphical depiction of this is presented in Fig. 6, showing that the negative across-subject correlation between the right SMA and right area MT ($r = -.55$, $p = .03$) during the *Past* condition (see Fig. 6) was not present across trials within a single participant in the study. Indeed, for most participants, the temporal (within-subject) correlations were positive, suggesting that these two regions are tightly coupled during the *Past* condition. A one-sample *t*-test is supportive of this notion, showing that these within-subject *r*-values (after an *r*-to-*z* transformation) are significantly greater than zero ($t(15) = 7.28$, $p < .0001$). It should be noted that these instances of Simpson's Paradox are relatively rare: of the 27 significant *as-fcMRI* relationships, only 2 of these produced a significant *ws-fcMRI* effect in the opposite direction, 17 produced a significant *ws-fcMRI* effect in the same direction, and 8 failed to produce a significant *ws-fcMRI* effect.

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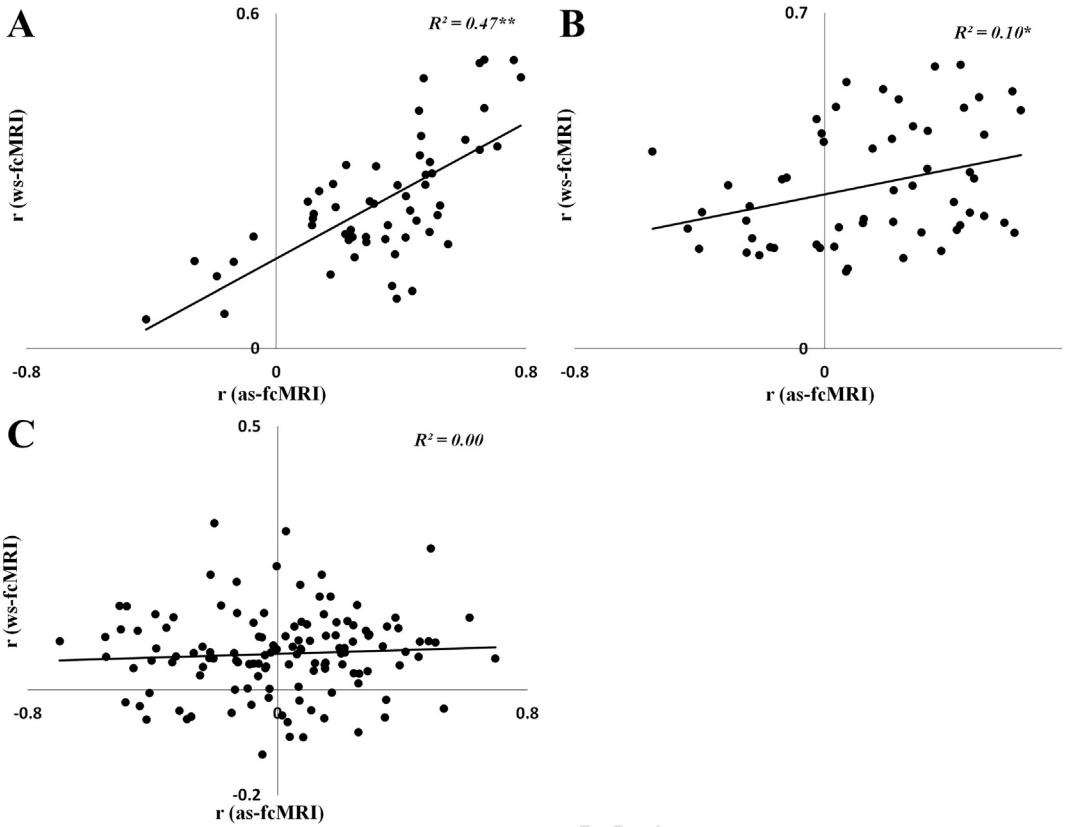

Fig. 5. The relationship between across-subject (*as-fcMRI*) and within-subject (*ws-fcMRI*) *r*-values between regions in: (A) the default-mode network; (B) the dorsal attention network; and (C) different networks. R^2 values are included on each plot (* = $p < .05$; ** = $p < .001$). *as-fcMRI* = across-subject task-related functional connectivity; *ws-fcMRI* = within-subject task-related functional connectivity.

3.3. Comparing as-fcMRI and ws-fcMRI within the PLS framework

One of the advantages of seed PLS as an fMRI analytic tool is the increased power afforded by its multivariate approach (McIntosh et al., 1996, 2004). However, as shown in our analyses thus far, a serious issue with *as-fcMRI* methods such as as-seed PLS is that they do not index the *temporal* interactions between brain regions. In this section, we compare the results of a standard as-seed PLS analysis with a novel analysis (ws-seed PLS) that combines the multivariate approach of PLS with temporally-derived within-subject correlations.

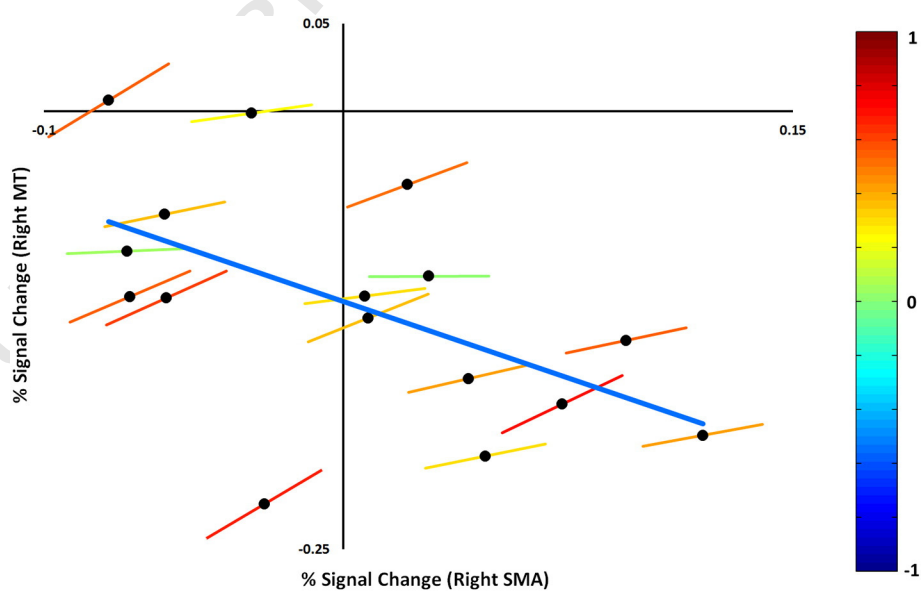

Fig. 6. Across-subject correlation (blue line) showing a negative relationship between each participant's mean percent signal change of the right SMA and area MT during the *Past* condition (black dots). Within-subject *r*-values are represented by lines intersecting each participant's mean. The colour of each line represents the strength of the correlation, as shown in the colour bar. All within-subject *r*-values are greater than zero. MT = middle temporal area; SMA = supplementary motor area. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

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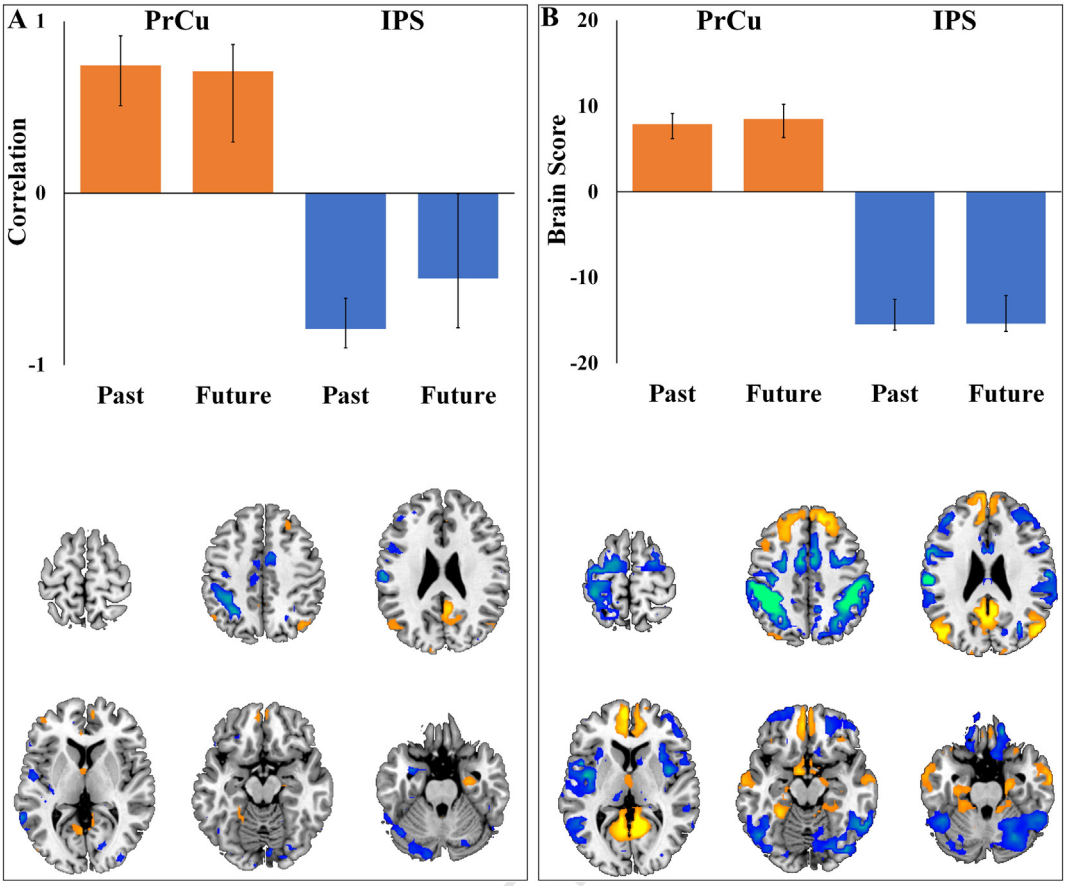

Fig. 7. Depiction of significant latent variables showing: (A) as-seed PLS correlation profile of precuneus and IPS seeds during the *Past* and *Future* conditions (top) and the associated singular image indicating regions showing reliable across-subject correlations with the seeds (bottom). Note that correlation profile indicates the strength of the across-subject relationship between signal in the seed region and brain scores for both conditions (B) Mean ws-seed PLS brain scores indicating differential *ws-fcMRI* for precuneus and IPS seeds during the *Past* and *Future* conditions (top) and the associated singular image showing brain regions expressing the effect of the latent variable (bottom). Note that brain scores indicate the average degree to which participants express the spatial *ws-fcMRI* connectivity pattern. For both plots, error bars are bootstrapped 95% confidence intervals. For both singular images, warm regions show increased functional connectivity to the precuneus, and cool colours show increased functional connectivity to the IPS, thresholded at a BSR ≥ +/− 2 ($p < .05$). BSR = bootstrap ratio; IPS = intraparietal sulcus; PrCu = precuneus; *ws-fcMRI* = within-subject task-related functional connectivity.

A non-rotated as-seed PLS analysis was performed resulting in a significant latent variable ($p < .001$) that identified regions showing differential across-subject correlations with the precuneus and IPS seeds during the *Past* and *Future* conditions. Fig. 7A shows that, during the *Future* and *Past* conditions, one set of regions (weighted positively in this latent variable) showed greater across-subject correlations with the precuneus than IPS seed while another set of regions (weighted negatively) showed the reverse effect.[9] The ws-seed PLS analysis also generated a significant latent variable ($p < .001$) identifying regions that reliably exhibited different *ws-fcMRI* with the precuneus and IPS during the Future and Past conditions — see Fig. 7B. As in the seed PLS analysis, regions showing greater within-subject correlations to the precuneus than the IPS seed were weighted positively, while those showing the opposite effect were weighted negatively.

To ascertain whether the regions connected to the precuneus and IPS belonged to the DMN or DAN, spatial maps of each network were downloaded from the online database Neurosynth (http://neurosynth.org/; Yarkoni et al., 2011). The DMN mask was defined as voxels showing reliable *rs-fcMRI* with the precuneus (4–56 12) and/or medial prefrontal cortex (−4 56–8), while the DAN mask was defined as voxels showing reliable *rs-fcMRI* with the left IPS (−36–40 44) and/or left supplementary motor area (−24–8 56). For both seed PLS analyses, significant clusters (thresholded at a bootstrap ratio of 2, corresponding to $p < .05$) that spatially overlapped with either network mask were deemed to belong to that network.

For the as-seed PLS analysis, many of the regions showing increased *as-fcMRI* to the precuneus seed (relative to the IPS) were in the DMN, including bilateral angular gyri, right hippocampus and medial prefrontal cortex — see Table 2. In addition, the precuneus seed also showed increased connectivity to some regions not typically associated with the DMN. For example, the left lateral frontal pole – a region that has previously shown to be strongly connected to the IPS at rest (Goulas, Uylings, & Stiers, 2012) – exhibited greater *as-fcMRI* with the precuneus than the IPS during the two experimental conditions. Regions showing greater *as-fcMRI* to the IPS compared to the precuneus were generally regions located in task-positive networks (e.g. left insula, bilateral cingulate, occipital and parietal cortices) as well as the default mode network (e.g., left medial prefrontal, temporal pole and middle temporal regions).

For the ws-seed PLS analysis, regions connected with the precuneus generally belonged to the default-mode network (e.g., bilateral medial prefrontal cortex, bilateral middle temporal gyri extending into temporal poles, bilateral medial temporal lobes and bilateral angular gyri). None of the clusters showing *ws-fcMRI* with the precuneus were within the DAN. Regions showing increased connectivity to the IPS included

[9] In seed-PLS analyses in which functional connectivity differs between seeds, conditions, groups etc., it is important to note that a voxel's salience is determined by a *difference in r-values*. As such, one should avoid describing, for example, all regions with positive saliences as necessarily being "functionally connected" to the precuneus (in this case). It could be the case that a positively weighted voxel has no relationship with the precuneus but is negatively correlated with the IPS.

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Table 2
Regions showing reliable differences in *as-fcMRI* to precuneus and IPS seeds in as-seed PLS analysis.

PrCu > IPS						*IPS > PrCu*					
Structure	MNI			BSR	Cluster size	Structure	MNI			BSR	Cluster size
	x	y	z				x	y	z		
Default mode network regions											
R PrCu	4	−56	16	7.25	251	L TP	−24	−4	−44	−5.62	59
R PHG	28	−12	−24	5.26	24	L OFC	−12	20	−20	−5.22	15
L AG	−44	−60	52	5.00	69	L MPFC	−12	44	8	−3.13	6
R AG	40	−72	44	4.74	70	L TP	−28	20	−36	−3.09	19
R MPFC	4	48	−8	4.74	96	L OFC	−44	28	−4	−2.89	20
L HypTh	−4	0	0	4.54	35						
R MFG	28	32	48	3.51	20						
L PHG	−24	−40	−12	3.30	22						
L MPFC	−8	64	12	3.13	6						
R CinG	8	36	20	2.99	15						
R MTG/TP	64	0	−20	2.91	5						
L PHG	−16	−12	−24	2.57	6						
Dorsal attention network regions											
L LFP	−40	48	8	3.75	19	L IPS	−40	−36	36	−6.47	273
L LOC	−48	−76	0	2.74	12	L MTG	−64	−48	0	−6.39	274
						L PoCG	−60	−16	24	−5.21	163
						R CinG	12	0	44	−4.38	36
						R CB	36	−36	−40	−4.36	13
						R LOC	24	−60	40	−3.74	15
						L Ins	−32	12	0	−3.67	12
						L PrCG	−52	4	16	−3.64	67
						R PoCG	64	−16	36	−3.61	16
						R PoCG	36	−32	40	−3.48	22
						L CinG	−8	−28	44	−3.44	49
						L PrCG	−24	−24	56	−3.35	13
						R MTG	60	−56	−4	−3.30	6
						L MFG	−40	40	28	−3.11	19
						L MTG	−48	−32	−4	−3.09	8
						R LOC	52	−68	−24	−3.07	24
Other regions											
L CB	−52	−60	−44	5.26	13	R CB	12	−60	−48	−5.34	29
L CB	−8	−44	−20	3.49	13	L OFC	−20	36	−20	−4.83	21
L CB	−44	−68	−52	3.34	11	R POC	44	−20	20	−4.24	13
R STG	52	−16	−8	2.96	13	R PHG	16	4	−40	−4.24	20
L OccP	−12	−96	24	2.83	9	R LinG	8	−84	−16	−3.57	13
L Med SFG	−4	32	40	2.70	6	R CB	24	−76	−28	−3.37	49
L PrCu	−4	−48	44	2.62	6	R Calcarine	20	−76	4	−3.37	15
R CB	4	−56	−36	2.59	7	R MOG	36	−88	4	−3.18	14
						L IFG	−40	44	−20	−3.04	7
						L MFG	−28	44	24	−2.85	7
						R MOG	36	−84	24	−2.84	8
						R LinG	20	−52	−12	−2.76	5
						R TP	48	12	−20	−2.76	6
						R PrCG	60	4	12	−2.71	5
						R ITG	48	−36	−28	−2.67	7

Note: Cluster size is in voxels. Bootstrap ratios of all clusters were greater than ±2 (roughly equivalent to a p-value of <.05), and had a spatial extent of at least 5 voxels (4 × 4 × 4 mm). AG = angular gyrus; *as-fcMRI* = across-subject functional connectivity; as-seed PLS = across-subject seed PLS; BSR = bootstrap ratio; CB = cerebellum; CinG = cingulate gyrus; HypTh = hypothalamus; IFG = inferior frontal gyrus; Ins = Insula; IPS = intraparietal sulcus; ITG = inferior temporal gyrus; L = left; LFP = lateral frontal pole; LinG = lingual gyrus; LOC = lateral occipital complex; Med SFG = medial superior frontal gyrus; MFG = middle frontal gyrus; MNI = Montreal Neurological Institute; MOG = middle occipital gyrus; MPFC = medial prefrontal cortex; MTG = middle temporal gyrus; OccP = occipital pole; OFC = orbitofrontal cortex; PHG = parahippocampal gyrus; POC = parietal operculum cortex; PoCG = postcentral gyrus; PrCG = precentral gyrus; PrCu = precuneus; R = right; STG = superior temporal gyrus; TP = temporal pole.

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many regions of the DAN, including lateral parietal, frontal and occipital regions. However, as shown in Table 3, a single cluster in the orbitofrontal cortex overlapped with the default-mode network mask.

To summarise, for both analyses, it was the case that regions correlating with a seed region generally belonged to the same functional network as the seed. There were, however, some cases in which a seed showed functional connectivity to a region generally associated with a network negatively correlated with the seed region at rest. These cases were more common for the as-seed than the ws-seed PLS analysis.

Next, we sought to determine if any regions identified in the previous PLS analyses exhibited a Simpson's Paradox effect in which the direction of the as-seed PLS effect is reversed in the ws-seed PLS analysis. That is, we assessed whether any voxels showing a precuneus > IPS effect for the as-seed PLS analysis showed an IPS > precuneus effect for the ws-seed PLS analysis (and vice-versa). In total, four clusters – shown in Fig. 8 – showed a Simpson's Paradox effect. Clusters in the left temporopolar and medial prefrontal cortex showed an IPS > precuneus effect for the as-seed PLS analysis, and the reverse effect for ws-seed PLS. Conversely, left lateral frontopolar and right angular gyrus clusters showed a precuneus > IPS as-seed PLS effect, and an IPS > precuneus ws-seed PLS effect. Interestingly, for regions exhibiting a Simpson's Paradox effect, the direction of the effects in the ws-seed PLS analysis agreed with findings from *rs-fcMRI* (as reported in the online database www.neurosynth.org).

Importantly, voxels showing a Simpson's Paradox were in the minority, and were far outnumbered by voxels showing effects in the same direction for both analyses. Of all voxels showing a reliable effect for the as-seed PLS analysis (i.e., with bootstrap ratios >2 or

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Table 3
Regions showing reliable differences in *ws-fcMRI* to precuneus and IPS seeds in ws-seed PLS analysis.

PrCu > IPS						*IPS > PrCu*					
	MNI						MNI				
Structure	x	y	z	BSR	Cluster size	Structure	x	y	z	BSR	Cluster size
Default mode network regions											
R PrCu	4	−52	8	25.74	425	R OFC	16	40	−24	−9.41	358
L MPFC	−4	56	4	11.57	1025						
L AG	−44	−76	28	11.41	143						
R AG	52	−68	20	8.86	131						
R MTG	64	−8	−20	7.66	80						
L PHG	−28	−40	−12	6.84	121						
L MTG	−64	−4	−20	5.59	106						
R OFC	32	32	−16	5.00	8						
R CB	4	−48	−48	4.63	15						
L OFC	−36	28	−20	3.58	14						
L TP	36	24	−32	3.22	10						
R CB	28	−88	−40	2.84	16						
L CB	−28	−80	−40	2.82	5						
L CB	−8	−48	−44	2.37	6						
Dorsal attention network regions											
						L IPS	−36	−44	44	−23.55	1468
						R MFG	28	0	56	−13.47	572
						R IPL	48	−32	48	−13.32	687
						R PrCu	8	−56	52	−7.46	64
						R FFG	36	−76	−20	−6.99	574
						L FFG	−40	−48	−20	−6.71	454
						L MFG	−40	40	24	−5.90	117
Other regions											
L caudate	−12	12	8	4.12	13	R CB	8	−68	−52	−6.87	34
R OccP	16	−92	28	3.86	10	L OFC	−20	40	−20	−5.66	100
						R Thalamus	16	−28	12	−5.39	39
L OccP	−8	−92	32	3.45	17						
R OFC	28	20	−24	3.26	6	L CB	−12	−68	−52	−4.92	25
						L Thalamus	−12	−24	8	−4.67	26
L IFG	−48	24	0	3.08	10						
L CB	−12	−56	−20	2.78	16	L PocG	−12	−44	60	−4.67	28
						R LinG	16	−84	−12	−4.64	7
						R OccP	12	−92	−4	−4.31	46
						R TP	32	8	−48	−4.05	7
						L putamen	−24	16	0	−3.83	5
						R caudate	12	12	0	−3.40	13
						L CinG	16	−20	36	−3.35	26
						L ITG	−40	−8	−48	−3.33	5
						R HesG	40	−20	8	−3.10	10
						R caudate	16	20	−8	−2.72	5
						L FFG	−32	−8	−40	−2.72	7
						L calcarine	−8	−80	8	−2.27	5

Note: Cluster size is in voxels. Bootstrap ratios of all clusters were greater than ±2 (roughly equivalent to a p-value of <.05), and had a spatial extent of at least 5 voxels (4 × 4 × 4 mm). AG = Angular Gyrus; CB = cerebellum; CinG = cingulate gyrus; FFG = fusiform gyrus; HesG = Heschl's gyrus; IFG = inferior frontal gyrus;IPL = inferior parietal lobule; IPS = intraparietal sulcus; L = left; LinG = lingual gyrus; MFG = middle frontal gyrus; MNI = Montreal Neurological Institute; MTG = middle temporal gyrus; OccP = occipital pole; OFC = orbitofrontal cortex; PoCG = postcentral gyrus; PrCu = precuneus; R = right; TP = temporal pole; *ws-fcMRI* = within-subject task-related functional connectivity; ws-seed-PLS = within-subject seed PLS.

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<−2), 60.3% showed effects in the congruent direction for the ws-seed PLS analysis, 36.5% showed no reliable ws-seed PLS effect and only 3.2% showed ws-seed PLS effects in the opposite direction (i.e., a Simpson's Paradox). The correspondence of the two analyses types was also reflected in a moderate across-voxel linear relationship between the bootstrap ratios of each analysis ($R^2 = .28$), although this relationship was substantially reduced when voxels within 4 mm of the perimeter of the seed region were not included in the regression ($R^2 = .23$).

The high proportion of voxels showing congruent effects for the two analyses is likely due to the seed regions belonging to distinct functional networks; as shown in Section 3.2, *as-fcMRI* and *ws-fcMRI* are both able to reliably distinguish the DMN and DAN networks. If this explanation is correct, then the proportion of voxels showing congruent as-seed PLS and ws-seed PLS effects should be dependent on the location of the seed regions. Additional analyses were performed to determine the proportion of voxels showing congruent and opposite (i.e., Simpson's Paradox) effects when the two seeds were both from the DMN. To this end, we computed new as-seed PLS and ws-seed PLS analyses on data from the *Future* and *Past* conditions using three pairs of seed regions: (i) left MPFC and right precuneus; (ii) left and right MTL; and (iii) left and right angular gyrus. Chi-squared tests were performed to test, for each of the additional pairs of analyses, if the number of significant *as-fcMRI* voxels showing congruent, incongruent or no reliable *ws-fcMRI* effects was different to the original analysis — this was the case for all three additional seed PLS analyses (all p's < .00001). As shown in Table 4, the percentage of significant as-seed PLS voxels showing incongruent ws-seed PLS effects was similar for all four sets of analyses, but the number of voxels showing congruent ws-seed PLS voxels dropped substantially in the additional analyses. This suggests that the degree to which as-seed PLS and ws-seed PLS effects cohere is contingent on whether the seed regions being tested from the same or distinct resting-state networks. The overall reduction in similarity between the two analysis types when seeds were from the same network was also

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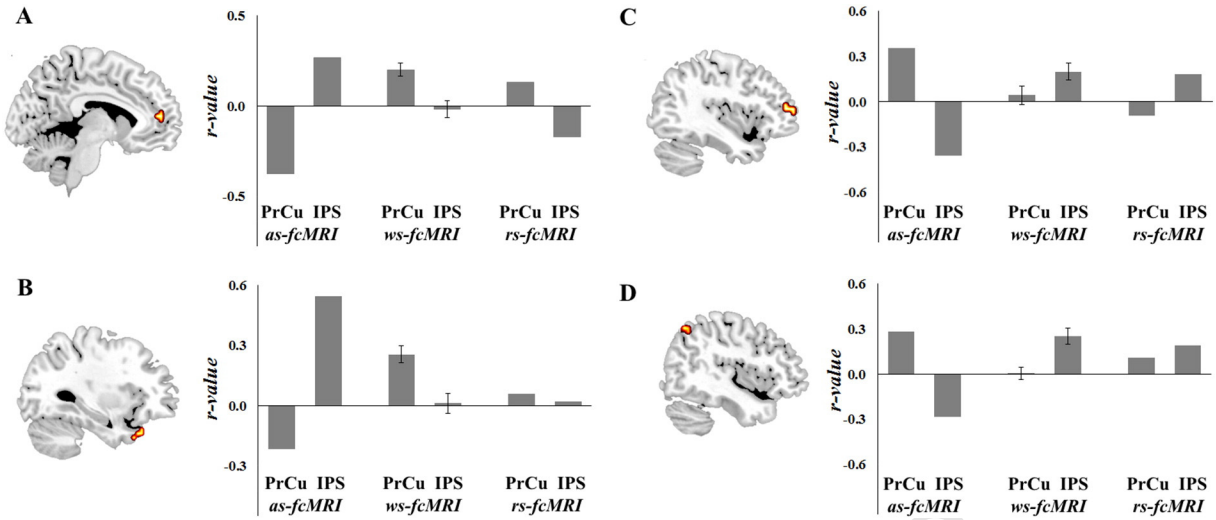

Fig. 8. Regions showing a Simpson's paradox effect in which the direction of an as-seed PLS effect is reversed in the ws-seed PLS analysis. Shown here are four clusters: (A) left medial prefrontal cortex (*xyz* −8 44 8); (B) left temporal pole (−36 24 −32); (C) left lateral frontal pole (−40 48 8); (D) right angular gyrus (36 −68 48). Plots show the corresponding correlation coefficients (averaged for the *Past* and *Future* conditions) between the seed regions (precuneus and IPS) and a representative voxel in that cluster for the as-seed PLS and ws-seed PLS analyses. Also shown are correlation coefficients obtained from *www.neurosynth.org* for a resting-state functional connectivity meta-analysis for the same seed and voxel (Yarkoni et al., 2011). Error bars for *ws-fcMRI* represent standard error of the mean within-subject *r*-value. *as-fcMRI* = across-subject task-related functional connectivity; IPS = intraparietal sulcus; PrCu = precuneus; *rs-fcMRI* = resting state functional connectivity; *ws-fcMRI* = within-subject task-related functional connectivity.

reflected in a reduction in the strength of the linear relationships between the bootstrap ratios of each analysis (see Table 4).

4. Discussion

Functional connectivity is investigated by computing correlations between BOLD signal from different brain regions or voxels. The nature of the correlations computed differs fundamentally between within- and across-subject analysis techniques: the former assesses temporal correlations within subjects while the latter assesses atemporal correlations across subjects. Because these two approaches both fall under the umbrella of "functional connectivity", it is essential to determine how closely they align with each other. The possibility of a Simpson's Paradox effect provides an a priori reason to suggest that two regions showing a reliable across-subject correlation will not necessarily show the same effect for within-subject correlations. The current study directly compared these two classes of functional connectivity during episodic simulation tasks, and showed that this is indeed the case under certain conditions.

Our analyses revealed a number of key findings. First, linear regressions showed that across-subject correlations between a seed region and voxels in the rest of the brain only predicted within-subject correlations to a weak-to-moderate degree. The precise strength of this relationship varied depending on the seed region chosen. For example, the two correlation types showed relatively high linear relationships when the seed region was the left angular gyrus for both *Future* and *Past* conditions. However, this relationship was considerably weaker for medial temporal lobe seeds and is likely due to the poor temporal signal-to-noise ratio in the medial temporal lobe (Bellgowan, Bandettini, van Gelderen, Martin, & Bodurka, 2006; Bellgowan, Buffalo, Bodurka, & Martin, 2009). This weak relationship between *as-fcMRI* and *ws-fcMRI* has an implication for studies that use *as-fcMRI* to assess task-related functional connectivity of regions in the medial temporal lobe (Barnett, O′Neil, Watson, & Lee, 2014; Burianova, McIntosh, & Grady, 2010; De Brigard, Nathan Spreng, Mitchell, & Schacter, 2015; McCormick, Moscovitch, Protzner, Huber, & McAndrews, 2010; McCormick, St-Laurent, Ty, Valiante, & McAndrews, 2013; McLelland, Chan, Ferber, & Barense, 2014; O′Neil et al., 2012; Paleja, Girard, Herdman, & Christensen, 2014; Robin et al., 2015; Sheldon & Levine, 2013; Söderlund, Moscovitch, Kumar, Mandic, & Levine, 2012), including our own (Hach et al., 2014; Martin et al., 2011; van Mulukom et al., 2013): It is questionable whether these *as-fcMRI* studies inform our understanding of "functional connectivity" as the term is generally used in the literature (i.e., as an index of a temporal, within-subject relationship), especially when the seeds are in the medial temporal lobes.

Next, we showed that both *as-fcMRI* and *ws-fcMRI* are generally able to dissociate the DMN and DAN, with correlations between regions of the same network being significantly greater than correlations between regions of different networks. A striking feature of the *as-fcMRI* correlation matrix was the greater variability in correlation coefficients between regions from different networks, and included a mixture of positive and negative correlations. Importantly, while the strength of the *ws-fcMRI* relationship between regions within a network were reliably predicted by the degree of *ws-fcMRI* between these regions, this was not the case between regions from different functional networks. This set of results are congruent with an important feature of many published seed PLS results: one seed shows a mixture of connections with regions in the same functional network in addition to regions from different resting-state functional networks. For example, Gerlach et al. (2011) demonstrated that seeds from the DMN (posterior cingulate) and the central executive network (bilateral dorsolateral prefrontal cortex) both show reliable *as-fcMRI* with a set of regions in both the DMN (e.g., medial prefrontal cortex, hippocampus) and task-positive networks (e.g., supplementary motor area, superior parietal lobule). The results of the current study suggest that such "between-network" functional connections— based on across-subject correlations—should be

Table 4
Percentage of voxels showing a reliable as-seed PLS effect and either a congruent, incongruent or no ws-seed PLS effect for different pairs of seed regions.

Seeds	ws-seed PLS effect			R^2
	Congruent	Incongruent	No effect	
Precuneus and IPS	60.29	3.20	36.51	.28 (.23)
Precuneus and MPFC	21.99	2.28	75.73	.08(.05)
L and R angular gyri	28.74	1.76	69.50	.12 (.09)
L and R MTL	15.13	5.77	79.11	.05(.02)

Note: Also shown are R^2 values indicating the linear relationship between voxel bootstrap ratios for each analysis. Values in parentheses are R^2 values when voxels within 4 mm of the seed regions are excluded. as-seed-PLS = across-subject seed partial least squares; L = left; R = right; ws-seed-PLS = within-subject seed partial least squares.

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interpreted cautiously: while *as-fcMRI* is a reasonably good predictor of *ws-fcMRI* between regions from the same network, there is no concordance between the two measures of functional connectivity between regions from different functional networks ($R^2 = 0.00$). An additional property of the correlation matrices generated was the existence of a Simpson's Paradox effect, showing that not only are across-subject correlations not necessarily mirrored in within-subject correlations, but that the two types of correlations can exhibit significant effects in opposite directions.

Finally, we compared the results of a standard seed PLS analysis (as-seed PLS) with those of a novel seed PLS approach based on within-subject correlations (ws-seed PLS). Seed regions used for these analyses were located in the DMN (precuneus) and DAN (IPS), and non-rotated seed PLS analyses were performed to determine which regions showed differential functional connectivity with these seeds during the episodic simulation tasks. The two analyses produced largely congruent results, with both analyses finding that the precuneus and IPS generally connect with other regions in their respective resting-state networks. However, there were some cases, primarily in the as-seed analyses, when a seed region showed increased functional connectivity with regions in a distinct functional network. In addition, four clusters of voxels showed a Simpson's Paradox effect, in which the direction of the as-seed PLS was reversed in the ws-seed PLS analysis. Importantly, the direction of the ws-seed PLS effect was always reflected in *rs-fcMRI* results obtained from www.neurosynth.com. These results have implications for how as-seed PLS results are interpreted: as-seed PLS analyses often show a region to be connected to areas outside its resting-state functional network, and this is interpreted as a case of a cognitive task inducing the coupling of regions that do not normally covary at "rest" (e.g. Robin et al., 2015; Spreng and Grady, 2010). An alternative—and simpler—explanation is that any difference in connectivity between the results of *rs-fcMRI* and as-seed PLS analyses are because the two approaches employ fundamentally different correlational techniques in their assessment of functional connectivity.

While the as-seed PLS and ws-seed PLS analyses yielded similar results, examination of the spatial maps produced by each method (see Fig. 7) shows that ws-seed-PLS produced more robust effects, suggesting that one explanation for the difference between *ws-fcMRI* and *as-fcMRI* is differential power to detect effects (i.e., *ws-fcMRI* has greater power). If this is the case, then voxels showing robust as-seed PLS effects should also show effects in the corresponding ws-seed PLS analysis. However, 37% of voxels showing a reliable as-seed PLS effect when precuneus and IPS were used as seeds failed to show an effect for the corresponding ws-seed PLS analysis. The difference in findings between the techniques was more striking for seed-PLS analyses involving DMN seeds. In these cases, 70–79% of voxels showing a reliable as-seed PLS effect did not show a reliable ws-seed PLS effect. The high proportion of voxels showing significant as-seed PLS effects and no reliable ws-seed PLS effect makes it unlikely that differences in these techniques are due to as-seed PLS being underpowered relative to ws-seed PLS. Instead, the differences between the two techniques are ultimately due to within- and across-subject correlations assessing fundamentally different sources of variance (Hamaker et al., 2005).

We propose that to validly investigate how resting-state networks are affected by cognitive tasks, task-related functional connectivity measures must be derived from methods that interrogate the temporal within-subject relationships between brain regions in the same way that resting-state methods do. A number of studies have adopted this within-subject approach, and have shown that cognitive tasks produce changes in the temporal relationship between regions within and between resting-state functional networks (e.g. Cocchi et al., 2013; Cole et al., 2014). Pertinent to the autobiographical tasks employed in the current study, Elton and Gao (2015) investigated how a range of cognitive tasks, including autobiographical memory retrieval, impact resting-state functional networks. Interestingly, they found that the autobiographical memory task produced less of a shift in the organisation of resting-state networks than did other, externally-directed cognitive tasks (e.g. N-back, no-go tasks). While this observation is relevant to the comparison of our *ws-fcMRI* results with *rs-fcMRI* networks, future research is required to address whether different cognitive tasks differentially affect the relationship between *as-fcMRI* and *ws-fcMRI*.

Given the differences observed in the current study, the immediate question that emerges is which class of functional connectivity analyses (*as-fcMRI* or *ws-fcMRI*) is best suited to investigate task-related functional connectivity? We agree with Friston (1994, 1995) that functional connectivity is best thought of as a measure of the temporal dynamical relationship between brain regions. Adopting *ws-fcMRI* techniques that are motivated by this definition has the advantage of allowing researchers to directly compare task-related connectivity findings to the large body of research on resting-state networks without the complication of using a fundamentally different correlational technique as would be the case if using *as-fcMRI*. In addition, as noted by Friston (1995), temporal correlations allow for the synthesis of fMRI functional connectivity results with those acquired from other modalities that employ temporal coherence as an index of functional connectivity (e.g. EEG/MEG), as has been done in a number of studies (Chang et al., 2013; Laufs, 2008; Liljeström et al., 2015; Tewarie et al., 2014). Another important advantage of *ws-fcMRI* is that it can determine how individual differences in connectivity relate to behaviour — something that is simply not possible using *as-fcMRI*, as this technique does not provide a measure of functional connectivity for *each* participant. Within the ws-seed PLS framework presented here, relating *ws-fcMRI* to behaviour is relatively simple: behavioural PLS analyses can be performed by relating behavioural performance (e.g. accuracy) to matrices containing within-subject correlation coefficients. Although beyond the scope of the present study, we explored the feasibility of this approach by using the precuneus and IPS seeds from the ws-seed PLS analyses to examine whether our current measures of *ws-fcMRI* were associated with task-performance. We found strong positive relationships between reaction time and the strength of connectivity of the precuneus with other regions in the DMN (e.g. medial prefrontal cortex, lateral temporal regions and angular gyrus—see Supplementary Materials for detailed results). Going forward, we believe that *ws-seed PLS* will prove a powerful tool for understanding how the connectivity of whole-brain networks underpins aspects of cognitive performance.

In conclusion, the current study has shown that while functional connectivity is a widely-used analytic tool in the field of fMRI research, the degree to which *as-fcMRI* and *ws-fcMRI* converge on the same findings varies depending on the seed region used and whether the seed and target voxels belong to the same or different functional networks. While the two correlational methods often produce similar covariance structures for regions within the same functional network, they show no agreement for regions between functional networks. Moreover, we have showed that although there is substantial congruence between the results yielded by the two methods in some cases, it can nevertheless be the case that *as-fcMRI* and *ws-fcMRI* produce reliable effects in opposite directions (i.e., a Simpson's Paradox). The results presented highlight the need to be cautious when interpreting across-subject correlations, as they are not necessarily informative about temporal interactions of brain regions within individual brains. Instead, these findings should be interpreted as a measure of how strongly across-subject variance in mean BOLD signal is shared between brain regions (i.e. as an index of individual differences). If the goal is to investigate task-related temporal relationships between brain regions, we recommend that *ws-fcMRI* should be utilized. Moreover this approach enables the direct comparison of results with *rs-fcMRI* and yields measures of connectivity strength for each participant that can be associated with behavioural performance, providing the opportunity to fully explore the richness of temporal task-related fMRI data and its relation to cognition.

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Appendix A. Supplementary data

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